

BA Farms — Master Feedlot Ration & Procurement Model

Least-Cost Linear Programming · Silage 27% DM · 95-Day Monsoon Cycle · Weekly Schedule
Sahiwal × Friesian crossbred | Faisalabad, Punjab | July 2026

Headline result

Item	Value
Feeds evaluated	18 inputs, all at supplied prices
Maize silage	27% dry matter, fully fermented (21+ days)
Purchase weight	165 kg
Days on feed	95 (ration re-scaled every 7 days)
Exit weight	264.9 kg
Total gain	99.9 kg
Average daily gain	1.05 kg/day
Total cost incl. minerals	Rs 31,969
ALL-IN COST PER KG OF GAIN	Rs 320
Feed-only cost per kg gain	Rs 285
Same plan, cool season (Oct–Mar)	Rs 261/kg at 1.2 kg/day

The optimiser selected six feeds: maize silage, maize grain, maize gluten feed 30%, toori, urea and limestone. Feed-only cost is Rs 285 per kg of gain; with the mineral and additive pack the all-in figure is **Rs 320 per kg**. All maize oil cakes, maize brans, corn gluten meal, rice polish, wheat bran, molasses, wanda and press mud are rejected on price. A separate scenario for green chari at Rs 8/kg appears in Section 10.

Objective used. The model minimises cost per kilogram of gain *subject to a minimum daily gain of 1.00 kg*. This floor matters: without it, the optimiser will buy cheap, slow growth that misses your 95-day exit window. Every figure in this report respects the timeline.

1. Price basis

Feed	Rs/kg	As purchased	Status
Maize silage	16.25	—	FIRM
Maize grain (makai)	55.00	Rs 2,200 / 40 kg	FIRM
Maize gluten feed 30% (Rafhan)	68.00	Rs 3,400 / 50 kg	FIRM
Cottonseed cake — mithi khal	80.00	Rs 4,000 / 50 kg	FIRM — bag size assumed 50 kg
Cottonseed cake — kori khal	74.00	Rs 3,700 / 50 kg	FIRM — bag size assumed 50 kg
Maize oil cake germ (Premier)	112.00	Rs 5,600 / 50 kg	FIRM — bag size assumed
Maize oil cake MOC (Rafhan)	122.00	Rs 6,100 / 50 kg	FIRM — bag size assumed
Maize bran (Rafhan)	66.33	Rs 1,990 / 30 kg	FIRM
Maize bran fibre (Premier)	62.67	Rs 1,880 / 30 kg	FIRM
Corn gluten meal 60%	239.80	Rs 11,990 / 50 kg	FIRM
Rice polish	65.00	Rs 2,600 / 40 kg	FIRM
Wheat bran (chokar)	73.33	Rs 2,200 / 30 kg	FIRM
Molasses (sheera)	60.00	Rs 60 / kg	FIRM
Urea	92.00	Rs 4,600 / 50 kg	FIRM
Toori (wheat straw)	28.00	—	FIRM
Wanda	100.00	—	FIRM
Press mud	5.00	—	FIRM
Limestone (chuna)	25.00	—	ESTIMATE

Khal bag size. If the khal bags are 40 kg rather than 50 kg, mithi becomes Rs 100/kg and kori Rs 92.50/kg. This changes the blended cost of gain by only about Rs 2, because the ration uses roughly 0.1–0.2 kg per head per day. **Mithi is the better buy** at either bag size: Rs 364/kg protein against Rs 389 for kori at 50 kg bags. Neither competes with maize gluten feed at Rs 227.

2. Feed value ranking

Feed	Rs/kg	Rs per Mcal	Rs per kg protein	Selected?
Press mud	5.00	16.7	278	no
Maize grain	55.00	18.4	696	YES
Toori (wheat straw)	28.00	20.0	903	YES
Maize silage (27% DM)	16.25	24.1	752	YES
Maize gluten feed 30%	68.00	27.0	227	YES
Rice polish	65.00	27.8	556	no
Cottonseed cake — mithi khal	80.00	29.1	364	no
Maize bran (Rafhan)	66.33	30.1	663	no
Maize bran fibre (Premier)	62.67	30.3	696	no
Cottonseed cake — kori khal	74.00	31.5	389	no
Wheat bran (chokar)	73.33	32.0	556	no
Molasses (sheera)	60.00	33.3	2000	no
Wanda	100.00	39.2	694	no
Maize oil cake germ (Premier)	112.00	39.4	589	no
Maize oil cake (Rafhan)	122.00	43.0	642	no
Corn gluten meal 60%	239.80	74.0	400	no
Urea	92.00	—	32	YES
Limestone (est.)	25.00	—	—	YES

Maize grain is the cheapest dense energy (Rs 18.4/Mcal); silage the cheapest bulk energy; urea the cheapest protein by far (Rs 32/kg CP, capped at the safety limit); maize gluten feed the cheapest real protein after urea (Rs 227/kg CP). **Press mud shows the cheapest energy on paper yet is still rejected** — its energy density is about 1.07 Mcal per kg of dry matter against the ~2.9 the ration needs. Under a fixed intake ceiling, density beats price.

3. Weekly ration schedule — adjust every 7 days

Kilograms as-fed per head per day. The additive pack (Section 5) is fed on top from week 2, at half rate in week 1.

Wk	Live wt (kg)	Silage	Maize grain	Gluten feed	Toori	Urea	Limestone	ADG	Rs/day
1	165–170	5.25	1.48	0.10	0.20	0.024	0.021	0.65	205
2	170–177	5.22	2.52	0.18	0.26	0.041	0.035	1.03	291
3	177–184	5.42	2.62	0.19	0.26	0.043	0.037	1.04	301
4	184–191	5.63	2.71	0.19	0.27	0.044	0.038	1.05	310
5	191–199	5.87	2.81	0.20	0.27	0.046	0.039	1.06	320
6	199–206	6.12	2.91	0.20	0.26	0.047	0.041	1.07	330
7	206–214	6.39	3.01	0.20	0.25	0.049	0.042	1.08	339
8	214–222	6.67	3.11	0.21	0.23	0.050	0.043	1.09	349
9	222–229	6.98	3.21	0.21	0.21	0.052	0.045	1.10	360
10	229–237	6.99	3.29	0.22	0.27	0.053	0.046	1.10	367
11	237–245	7.33	3.40	0.22	0.24	0.055	0.047	1.11	378
12	245–252	7.68	3.50	0.21	0.20	0.056	0.049	1.12	388
13	252–260	7.73	3.58	0.23	0.25	0.058	0.050	1.12	396
14	260–265	8.12	3.69	0.23	0.20	0.059	0.051	1.13	406

Week 1 adaptation. Days 1–3: silage only plus 0.2 kg toori, no grain and no urea, unlimited clean water. Days 4–7: grain at about 60% of full rate with urea introduced gradually. Day 8: move to the week-2 ration.

4. Costs and procurement per head

Item	kg per head	Rs/kg	Cost per head	Per 100 head	% of feed
Maize silage	632.8	16.25	Rs 10,283	Rs 1,028,300	36.8%
Maize grain	277.4	55.00	Rs 15,257	Rs 1,525,700	54.6%
Maize gluten feed 30%	18.6	68.00	Rs 1,265	Rs 126,480	4.5%
cgm	0.0	239.80	Rs 0	Rs 0	0.0%
Toori	22.9	28.00	Rs 641	Rs 64,120	2.3%
Urea	4.5	92.00	Rs 414	Rs 41,400	1.5%
Limestone	3.9	25.00	Rs 98	Rs 9,750	0.3%
Feed subtotal	—	—	Rs 27,958	Rs 2,795,750	100%

Component	Per head	Per 100 head	Per kg gain
Feed ingredients	Rs 27,958	Rs 2,795,750	Rs 280
Minerals, vitamins, buffers, binder	Rs 4,012	Rs 401,150	Rs 40
TOTAL	Rs 31,969	Rs 3,196,900	Rs 320

Week	Weight (kg)	Gain	Cost Rs	Rs/kg gain	Cumulative
1	165–170	4.6	1,435	315	Rs 1,435
2	170–177	7.2	2,040	283	Rs 3,475
3	177–184	7.3	2,105	289	Rs 5,580
4	184–191	7.4	2,172	295	Rs 7,752
5	191–199	7.4	2,239	302	Rs 9,991
6	199–206	7.5	2,307	308	Rs 12,298
7	206–214	7.6	2,376	314	Rs 14,674
8	214–222	7.6	2,446	321	Rs 17,120
9	222–229	7.7	2,517	327	Rs 19,637
10	229–237	7.7	2,572	334	Rs 22,209
11	237–245	7.8	2,644	340	Rs 24,853
12	245–252	7.8	2,718	347	Rs 27,571
13	252–260	7.8	2,772	354	Rs 30,343

Week	Weight (kg)	Gain	Cost Rs	Rs/kg gain	Cumulative
14	260–265	4.5	1,626	360	Rs 31,969
Total	165–265	100	31,969	320	Rs 31,969

5. Mineral, vitamin and additive pack

Additive	g/head/day	Rs/kg	Rs/day	95-day total	Purpose
Salt (NaCl)	45	30	1.3	Rs 124	Sodium; drives water intake in heat
Trace mineral + vitamin premix	28	400	11.2	Rs 1,025	Zn, Cu, Se, Co, I, Mn + Vit A/D/E
Sodium bicarbonate	120	150	18.0	Rs 1,647	Rumen buffer — acidosis protection
Toxin binder	15	600	9.0	Rs 824	Aflatoxin control in monsoon
Ammonium chloride	12	350	4.2	Rs 384	Urinary calculi prevention
TOTAL	—	—	43.8	Rs 4,003	≈ Rs 40 per kg of gain

6. Forced inclusion of press mud

Press mud forced (kg/day)	Achievable ADG	Rs per kg of gain	Change vs optimal
0	1.09 kg/day	Rs 285	—
1	1.03 kg/day	Rs 338	+Rs 53 (+19%)
2	0.99 kg/day	Rs 466	+Rs 181 (+64%)
3	0.94 kg/day	Rs 612	+Rs 327 (+115%)
4	0.87 kg/day	Rs 742	+Rs 457 (+160%)

Press mud is your competitors' key ingredient and it is nearly free at Rs 5/kg, yet **every kilogram makes you poorer**. One kilogram raises the cost of gain by Rs 120 and pulls daily gain from 1.09 to 1.05. Three kilograms more than doubles the cost of gain. Mud occupies stomach space that must then be won back with expensive dense energy, and it lengthens days on feed — multiplying labour, mortality risk and interest. **Do not use press mud.**

7. Forced inclusion of wanda

Wanda forced (kg/day)	Achievable ADG	Rs per kg of gain	Change vs optimal
0	1.09 kg/day	Rs 285	—
1	1.05 kg/day	Rs 365	+Rs 80 (+28%)
2	1.00 kg/day	Rs 452	+Rs 167 (+59%)
3	—	INFEASIBLE	Exceeds stomach capacity

Wanda at Rs 100/kg is expensive in both directions. One kilogram per day raises the cost of gain by Rs 123; two kilograms by Rs 211. At three kilograms the binding problem is no longer cost but volume — wanda is bulky relative to its energy and no longer fits alongside the required roughage. If you keep wanda for supplier or convenience reasons, hold it to about 0.5 kg/day and treat the extra as a deliberate convenience cost.

8. Silage dry matter — sensitivity

Silage is priced per kilogram as fed, so a wetter silage means you are buying more water and less feed. Your silage is taken as 27% DM.

Silage DM %	Silage fed @220 kg	ADG	Rs per kg gain	vs 27% DM
22%	4.81 kg	1.00	Rs 306	+21
24%	7.51 kg	1.09	Rs 298	+13
26%	7.23 kg	1.09	Rs 289	+4
27%	6.96 kg	1.09	Rs 285	—
28%	6.71 kg	1.09	Rs 282	-3
30%	6.26 kg	1.09	Rs 275	-10
33%	5.69 kg	1.09	Rs 267	-18

Each percentage point of dry matter is worth roughly Rs 4–5 per kg of gain. Cutting the crop at the correct maturity — the milk-to-dough stage, around one-half milk line — rather than early for tonnage is worth about Rs 30 per kg of gain between a 22% and a 30% silage. **Verify your DM with the microwave test at feed-out:** weigh 100 g of chopped silage, microwave in two-minute bursts with a cup of water beside it, reweigh until the weight stops falling. The final weight in grams is your DM percentage.

9. Sensitivity — which price matters

Maize grain

Price (Rs/kg)	44.0	49.5	55.0	60.5	68.8	82.5
Cost per kg gain	Rs 253	Rs 269	Rs 285	Rs 301	Rs 325	Rs 362

Maize silage (27% DM)

Price (Rs/kg)	13.0	16.2	19.5	24.4
Cost per kg gain	Rs 265	Rs 285	Rs 304	Rs 324

Maize gluten feed 30%

Price (Rs/kg)	54.4	68.0	85.0	102.0
Cost per kg gain	Rs 283	Rs 285	Rs 288	Rs 290

Cottonseed cake — mithi khal

Price (Rs/kg)	56.0	80.0	112.0
Cost per kg gain	Rs 283	Rs 285	Rs 285

Urea

Price (Rs/kg)	73.6	92.0	138.0
Cost per kg gain	Rs 284	Rs 285	Rs 287

Maize grain dominates. A 50% price rise takes the cost of gain from Rs 290 to well over Rs 350. Silage is second. Maize gluten feed, cottonseed cake and urea barely register because they are used in small quantities. Ration refinement is now largely exhausted; **your remaining leverage is procurement** — see Section 12.

10. Scenario — green chari available at Rs 8/kg

This section assumes fresh green chari (sorghum/maize fodder) can be bought at Rs 8 per kilogram as fed. Chari is roughly 22% dry matter, so it is mostly water; it substitutes for silage, not for grain. Inclusion is capped at 30% of dry matter.

Measure	Without chari	With chari at Rs 8
Blended cost per kg gain	Rs 285	Rs 252
Average daily gain	1.09 kg	1.09 kg
Silage fed at 220 kg	6.96 kg	1.21 kg
Chari fed at 220 kg	—	7.02 kg
95-day cycle exit weight	264.9 kg	261.6 kg
95-day all-in cost per kg gain	Rs 320	Rs 280

Chari at Rs 8 saves about Rs 33 per kg of gain, roughly Rs 3,300 per head over the cycle, or Rs 330,000 across 100 head. It displaces most of the silage while daily gain is unchanged at 1.09 kg, because chari and silage do the same job and chari is cheaper per unit of dry matter at this price.

Chari price (Rs/kg)	Cost per kg gain	ADG	Chari fed @220kg	Silage fed @220kg
Rs 4	Rs 226	1.09	7.02 kg	1.21 kg
Rs 6	Rs 239	1.09	7.02 kg	1.21 kg
Rs 8	Rs 252	1.09	7.02 kg	1.21 kg
Rs 10	Rs 265	1.09	7.02 kg	1.21 kg
Rs 12	Rs 277	1.09	7.02 kg	1.21 kg
Rs 15	Rs 285	1.09	0.00 kg	6.96 kg

Chari stops being worth buying above about **Rs 14 per kilogram** — at Rs 15 the model reverts entirely to silage. Below that it enters at its maximum permitted inclusion.

Two cautions. First, chari is seasonal — cutting generally runs from roughly June through September and again in spring, so it cannot replace silage year-round; silage exists precisely to carry the same crop through the months chari is unavailable. Second, and more important: if the minimum-gain floor is removed, the optimiser will feed 16 kg of chari, drop maize almost entirely and settle at 0.76 kg/day because that is the cheapest cost per kilogram. That ration would exit at about 237 kg and miss your 95-day window. **Cheap bulky feed must always be paired with a minimum daily gain constraint**, which is how this report is built.

10B. Minimum-gain floor — why it is set at 1.00 kg/day

The floor is a safety net against cheap bulky feeds, not a performance target. Minimising cost per kilogram of gain already pushes gain as fast as it is economically worth going. Raising the floor does not make animals grow faster; it forces the model to buy ultra-dense expensive feed to hit a number, which is why cost explodes:

Floor (kg/day)	Rs per kg gain	Average ADG	Exit weight in 95 days	Verdict
0.95	Rs 285	1.09	268 kg	no benefit over 1.00
1.0	Rs 285	1.09	268 kg	RECOMMENDED — costs nothing, blocks the trap
1.05	Rs 290	1.09	269 kg	+Rs 5 for 0.5 kg more exit weight
1.1	Rs 349	1.11	271 kg	+Rs 64 for 2 kg — poor value
1.15	Rs 466	1.14	274 kg	+Rs 181 for 5 kg — very poor
1.2	Rs 664	1.18	277 kg	+Rs 379 for 8 kg — indefensible

Raising the floor from 1.00 to 1.20 would cost roughly **Rs 42,000 extra per head** to gain about 8 kg more exit weight — thousands of rupees per additional kilogram. The floor is therefore set at 1.00, where it costs nothing and still prevents the optimiser from selecting slow, cheap growth when a bulky feed such as chari is available.

10C. Dry-matter scenarios — 27%, 28%, 29% and 30% silage

Silage is bought by weight as fed, so a wetter silage means buying more water. All other inputs held constant, monsoon conditions.

Silage DM	Rs per kg gain	Average ADG	Exit in 95 days	Silage fed @220 kg
27%	Rs 285	1.09	268 kg	6.96 kg
28%	Rs 282	1.09	268 kg	6.71 kg
29%	Rs 278	1.09	268 kg	6.48 kg
30%	Rs 275	1.09	268 kg	6.26 kg

Costs in this table are **blended feed-only** per kilogram of gain, on the same basis as Section 10D — ingredients solved at five body weights and averaged, excluding the mineral pack and the adaptation week. Add roughly Rs 35–40 per kilogram to obtain the all-in budgeting figure.

Roughly **Rs 3–4 per percentage point of dry matter**. Moving from 27% to 30% is worth about Rs 10 per kg of gain, near Rs 1,000 per head. Daily gain is unaffected — only cost changes, because the animal simply eats a larger as-fed quantity of a wetter silage to obtain the same dry matter. Confirm your figure with the microwave test: weigh 100 g of chopped silage, microwave in two-minute bursts with a cup of water beside it, reweigh until the weight stops falling; the final weight in grams is the dry matter percentage.

10D. Intake scenarios — what happens if the animals eat more

Every figure elsewhere in this report assumes cattle consume 2.7% of body weight in dry matter, declining to 2.5% as they grow, reduced a further 10% for monsoon heat. This is a conservative assumption and it is **the single most influential number in the model**. Well-managed feedlot cattle can reach 3.0–3.2%. The following table shows what is on the table if you do.

Intake assumption	27% DM	28% DM	29% DM	30% DM
2.7% to 2.5% (current model)	Rs 285 1.09 kg/d exit 268 kg	Rs 282 1.09 kg/d exit 268 kg	Rs 278 1.09 kg/d exit 268 kg	Rs 275 1.09 kg/d exit 268 kg
2.9% to 2.7%	Rs 274 1.22 kg/d exit 281 kg	Rs 271 1.22 kg/d exit 281 kg	Rs 267 1.22 kg/d exit 281 kg	Rs 264 1.22 kg/d exit 281 kg
3.1% to 2.9%	Rs 266 1.33 kg/d exit 291 kg	Rs 263 1.33 kg/d exit 291 kg	Rs 260 1.33 kg/d exit 291 kg	Rs 257 1.33 kg/d exit 291 kg

How to read each cell. Every cell contains three numbers stacked in this order: (1) the **blended feed-only cost per kilogram of live-weight gain** in rupees; (2) the **average daily gain** in kg per day; (3) the **exit weight after 95 days** starting from 165 kg. For example the cell reading Rs 257 / 1.33 kg/d / exit 291 kg under the 30% DM column means: if your silage tests at 30% dry matter and your animals eat 3.1% of body weight in dry matter, the least-cost ration costs Rs 257 in feed ingredients per kilogram gained, the animal averages 1.33 kg/day, and reaches about 291 kg after 95 days.

Important — this is not the same basis as the Rs 320 headline figure. The numbers in this table are **feed-only**: they count ingredients solved at five body weights (160, 190, 220, 250, 280 kg) and averaged. They exclude the mineral, vitamin, buffer and binder pack (Rs 43.8 per head per day, about Rs 40 per kg of gain) and they exclude the week-1 adaptation period, during which the animal gains only about 0.65 kg/day while still eating. The **all-in** figure for the same 3.1% intake scenario, simulated day by day across all 95 days with the additive pack included, is approximately Rs 303 per kilogram of gain. Use the feed-only figures to compare scenarios against each other, because they isolate the variable being tested; use the all-in figures for budgeting and margin calculation, because that is what you actually spend.

Every cell except the baseline is hypothetical. The 2.9% and 3.1% intake rows are not measurements of your animals, and the 28%, 29% and 30% dry-matter columns are not measurements of your silage. They show what the model produces *if* those values hold. Until both are verified, the only cell describing your farm as it currently stands is the top-left: 2.7% intake at 27% dry matter, Rs 285 feed-only and Rs 320 all-in.

Intake dominates everything else in this report. Moving from 2.7% to 3.1% intake at 27% DM lowers the cost of gain from Rs 285 to Rs 266, raises average daily gain from 1.09 to 1.33, and lifts the 95-day exit weight from 268 kg to **291 kg — 23 kg more animal in the same time, at a lower cost per kilogram**. By comparison, improving silage dry matter from 27% to 30% is worth Rs 10. Intake is worth roughly twice as much as the entire dry-matter question, and it costs nothing to find out.

This also explains reports of 1.4 kg/day from other operators. At 3.1% intake this model reaches 1.33 kg/day, and in cool weather it reaches the same figure at ordinary intake. Such claims are therefore plausible for well-managed cattle in cool conditions or at heavier weights — though many farm claims arise from gut-fill error, since an animal weighed after feeding and drinking can register 15–20 kg heavier than the same animal weighed before. Always weigh at the same time of day, before the morning feed.

Correction to earlier terminology. The ceiling of about 1.2 kg/day quoted in earlier versions of this analysis was described as a genetic limit. That was inaccurate. It is an **intake-and-price ceiling** — a consequence of the assumed intake and of what the model is willing to pay for dense feed — not a biological limit of the animal. Genetics never binds in this model.

Cool season, for comparison:

Intake assumption	Cool season, 27% DM	Cool season, 30% DM
2.7% to 2.5% (current model)	Rs 257 1.33 kg/d exit 291 kg	Rs 249 1.33 kg/d exit 291 kg
2.9% to 2.7%	Rs 253 1.35 kg/d exit 293 kg	Rs 248 1.35 kg/d exit 293 kg
3.1% to 2.9%	Rs 249 1.35 kg/d exit 293 kg	Rs 248 1.35 kg/d exit 293 kg

In cool weather the model reaches 1.33 kg/day even at ordinary intake — heat alone is costing roughly 0.24 kg/day and about Rs 28 per kg of gain. Combined with the maize harvest price, this is the strongest argument for starting cycles in September–October.

10E. How to measure intake — the highest-value test available to you

Step	Action
When	From day 14 at the earliest, ideally day 21. Intake is genuinely depressed during transport recovery and rumen adaptation, so testing earlier measures the adaptation rather than the animal.
Days 1–3	Offer silage ad lib as part of the adaptation protocol. This is correct practice, not a test.
Method	Weigh what goes into the khurli and weigh the refusals the next morning, at the same time each day.
Target	2–5% refusal means they ate to appetite. A bunk licked completely clean with zero refusal means they were limited by what you offered, not by appetite.
If clean	Increase the offer by 5% and repeat until a small refusal appears. That point is their true intake.
Duration	Five to seven consecutive days, then average. Do not test during a heat spike or immediately after a weather change.
Then	Report the figure and the model can be re-solved at the true intake — potentially worth Rs 19 per kg of gain and 23 kg of extra exit weight.

11. Break-even prices for feeds not currently used

The maximum price at which each feed would enter the least-cost ration. If offered below these prices, buy and the ration should be re-optimised.

Candidate feed	Buy below	Season	Comment
Cull potato	Rs 14.4/kg	Jan–Mar (potato harvest)	Very likely below this in season — worth chasing
Whole cottonseed	Rs 80.9/kg	Oct–Jan (ginning season)	Ginning-season price often near this

Candidate feed	Buy below	Season	Comment
Corn steep liquor	Rs 43.5/kg	Year-round (Rafhan, liquid)	Ask Rafhan — liquid streams often priced to move
Hydrol	Rs 24.0/kg	Year-round (Rafhan, liquid)	Ask Rafhan; must beat Rs 24
Broken rice (kanki)	Rs 65.5/kg	Oct–Dec (rice milling)	Possible at mill gate in season
Bakery waste	Rs 78.1/kg	Year-round	Often available well below this — high priority
DDGS	Rs 100.1/kg	Year-round if distillery in range	Worth a quote
Wheat grain	Rs 76.9/kg	Apr–May (harvest)	Only if wheat falls below maize
Soybean meal	Rs 100.2/kg	Imported, year-round	Will not reach this price — ignore
Maize starch	Rs 86.5/kg	Year-round (Rafhan)	Food-grade exceeds this; only waste/off-spec qualifies

Wet feeds — potato, chari, wet pulp, press mud — are mostly water and substitute for silage, never for grain. Introduce gradually, cap inclusion around 25–30% of dry matter, and always keep a minimum-gain constraint. Potato must be chopped to prevent choking and must not be fed green or sprouted.

12. Bulk purchasing calendar

Feed	Cheapest window	Storage / action
Maize grain	Oct–Nov and May–Jun	Your largest cost line. Buy the full cycle requirement at harvest; dry below 14% moisture and store sealed on pallets. Highest-value action available to you.
Maize silage	Oct–Nov, also Apr–May	Make or buy a year in one go; a sealed bunker keeps 12 months. Cut at milk-to-dough stage, not early — see Section 8.
Toori	Apr–May (wheat harvest)	Buy the whole year in April; stores indefinitely under cover. Only ~19 kg per head per cycle needed.
Cottonseed cake / whole cottonseed	Oct–Jan (ginning)	Buy 2–3 months at a time; oil content means it goes rancid in heat.
Molasses	Nov–Mar (crushing)	Only worth buying under Rs 25/kg for this ration; currently Rs 60 and excluded.
Rice polish / broken rice	Oct–Dec (milling)	High oil, goes rancid within weeks in heat. Buy small and fresh.
Cull potato	Jan–Mar	Highly perishable — buy only what is fed within days.
Green chari	Jun–Sep and Mar–May	Seasonal only. See Section 10.
Rafhan byproducts	Year-round, stable	Negotiate volume contract rather than timing the market.
Urea	Avoid Nov–Dec and Jun	Fertiliser demand spikes at sowing. Only 4.5 kg per head per cycle needed. Store locked and dry.

The single highest-value decision in this document. Maize grain is roughly half your feed bill and moves from about Rs 44 at harvest to Rs 82 in the off-season. Buying the cycle requirement at the October–November harvest rather than summer spot is worth on the order of **Rs 700,000 per cycle across 100 head** — more than every ration adjustment in this report combined. Starting cycles in September–October also captures cool-season performance (Rs 261/kg at 1.2 kg/day versus Rs 285 at 1.09 in monsoon).

13. Model verification

Check	Supplied	Required / limit	Status
Solver status	HiGHS Status 7	Optimal	PASS
Dry matter intake	5.148 kg	≤ 5.148 kg	BINDING
Metabolisable energy	15.0439 Mcal	≥ 15.0439 Mcal	BINDING
Crude protein	0.6178 kg	≥ 0.6178 kg	BINDING
Calcium : phosphorus	1.69 : 1	≥ 1.5 : 1	satisfied

Intake, energy and protein are all exactly binding — the expected signature of a correct least-cost solution — and the calcium-to-phosphorus ratio clears its minimum. No constraint is violated.

14. Health risks and mandatory warnings

Urea poisoning — can kill within hours. The lethal dose for unadapted cattle is roughly 0.3–0.5 g per kg body weight; this ration feeds about half of that, so the margin is real but not generous. Dissolve urea in water or molasses and mix uniformly through the whole batch — never feed dry, never allow a lump. Introduce only from day 4 and build up. If animals go off feed for more than a day, restart at a low rate. Signs: staggering, tremors, bloat, frothing, laboured breathing. Store locked away from animals.

Urinary calculi (“water belly”). This is a grain-based, phosphorus-rich ration. Without limestone the calcium-to-phosphorus ratio collapses to about 0.29:1 against a 1.5:1 minimum, and male calves form stones that block the urethra and kill — usually the biggest, best-eating animals. Limestone and ammonium chloride are not optional. Ensure constant water access.

Acidosis and bloat. The 40% roughage floor and 0.2 kg of long toori are deliberate protection — do not cut them to save money. Feed as a properly mixed total ration; pouring grain on top of silage lets dominant animals eat the grain and crash. Sodium bicarbonate buffers the rumen and matters more in heat. Never let animals go hungry then feed a full grain ration.

Silage must be fully fermented before feeding. Fermentation is not complete until about day 21, when pH settles at 3.8–4.2. Feeding at day 7–14 risks retained nitrates, clostridial and listeria growth, and poor palatability — and opening the face early admits air that disrupts the rest of the pit. Feed out at least 30 cm of face per day in hot weather to prevent aerobic spoilage.

Aflatoxin. Monsoon humidity breeds mould in stored maize grain and cottonseed products, and this ration is maize-heavy. Buy in smaller lots during monsoon, store off the floor with airflow, reject musty or caked material, feed a toxin binder throughout. Note the tension with Section 12: bulk buying at harvest saves money but raises storage risk — invest in dry, ventilated, pallet-based storage before buying a full cycle of grain.

Disease. Haemorrhagic septicaemia is the classic monsoon killer of cattle in Punjab and can kill healthy animals within hours; vaccinate the whole herd before or at monsoon onset. Vaccinate for foot and mouth disease. Deworm on arrival. Keep pens drained — a lame animal stops eating, which ends its gain. Quarantine new arrivals for the first week.

15. Monsoon heat management

Measure	Detail	Why
Night feeding	60–70% of the ration between 6 pm and 8 am	Fermentation heat peaks 4–6 hours after feeding
Water	40–60 L/head/day; ~1 ft trough per 8–10 head; clean daily	Intake doubles in heat; cattle refuse fouled water
Shade	20–40 sq ft per head, roof 10–12 ft	Direct sun sharply cuts intake
Buffer	Sodium bicarbonate 120 g/head/day	Panting animals lose CO ₂ and buffering capacity
Electrolytes	Extra salt; adequate potassium	Sweating depletes sodium and potassium
Handling	Weigh and move only in early morning	Handling in peak heat can cause collapse

16. Scenario supplement — complete weekly schedules

Summary of all four plans

Plan	Exit weight	Total gain	Avg ADG	Total cost/head	All-in Rs/kg gain
Baseline (intake 2.7%→2.5%)	265 kg	100 kg	1.05	Rs 31,969	Rs 320
A — intake 2.9%→2.7%	277 kg	112 kg	1.18	Rs 34,792	Rs 310
B — intake 3.1%→2.9%	289 kg	124 kg	1.31	Rs 37,597	Rs 303
C — green chari at Rs 8	265 kg	100 kg	1.05	Rs 28,781	Rs 288

Read these two ways. Chari is the cheapest per kilogram of gain (Rs 288) but produces the same 100 kg of gain as the baseline — it saves money on the same output. Higher intake costs slightly more per kilogram than chari but **produces materially more animal in the same 95 days**: Scenario B yields 124 kg of gain against the baseline's 100 kg, a 24% increase in output from the same pen, the same labour and the same 95 days. If your selling price per kilogram exceeds your cost of gain, more output is worth more than a lower unit cost.

All three scenarios are hypothetical until verified. Scenarios A and B require confirming that your animals actually eat more than the conservative assumption — see the measurement protocol in the main report. Scenario C requires chari to be available at Rs 8/kg, which is currently uncertain as it is not the cutting season.

16A. Scenario A — intake 2.9% declining to 2.7% of body weight

This scenario assumes cattle consume 2.9% of body weight in dry matter at 160 kg, declining to 2.7% at 280 kg, before the 10% monsoon reduction. This is a realistic figure for well-managed feedlot cattle and is roughly 7% above the conservative assumption used in the main report. The ration composition is unchanged — the animal simply eats more of the same mix.

Measure	Value
Purchase weight	165 kg
Days on feed	95
Exit weight	277.2 kg
Total gain	112.2 kg
Average daily gain	1.18 kg/day
Feed cost per head	Rs 30,785
Minerals and additives	Rs 4,007
Total cost per head	Rs 34,792
ALL-IN COST PER KG OF GAIN	Rs 310

Weekly ration — kilograms as fed per head per day

Wk	Live wt (kg)	Silage	Maize grain	Gluten feed	Toori	Urea	Limestone	ADG	Rs/day	Rs/week
1	165–170	5.57	1.59	0.11	0.24	0.026	0.022	0.65	212	1,486
2	170–178	5.77	2.72	0.18	0.23	0.044	0.038	1.16	311	2,177
3	178–186	5.95	2.83	0.20	0.25	0.046	0.040	1.17	322	2,254
4	186–194	6.41	2.96	0.19	0.20	0.048	0.041	1.19	335	2,347
5	194–203	6.65	3.08	0.20	0.21	0.050	0.043	1.20	346	2,425
6	203–211	6.90	3.19	0.21	0.22	0.052	0.045	1.21	358	2,505
7	211–220	7.16	3.31	0.22	0.23	0.053	0.046	1.22	369	2,586
8	220–228	7.45	3.43	0.22	0.23	0.055	0.048	1.23	381	2,668
9	228–237	7.76	3.55	0.23	0.22	0.057	0.049	1.24	393	2,751
10	237–246	8.09	3.67	0.23	0.20	0.059	0.051	1.25	405	2,835
11	246–254	8.11	3.77	0.25	0.28	0.061	0.053	1.25	415	2,903
12	254–263	8.47	3.89	0.25	0.25	0.063	0.054	1.26	427	2,987
13	263–272	8.87	4.01	0.25	0.21	0.065	0.056	1.27	439	3,073
14	272–277	8.92	4.12	0.27	0.28	0.066	0.057	1.27	449	1,795

Total feed consumed and cost per head over the 95-day cycle

Item	kg per head	Rs/kg	Cost per head	Per 100 head	% of feed cost
Maize silage	704.1	16.25	Rs 11,441	Rs 1,144,100	37.2%
Maize grain	305.9	55.00	Rs 16,822	Rs 1,682,200	54.6%
Maize gluten feed 30%	19.8	68.00	Rs 1,349	Rs 134,900	4.4%
Toori	21.7	28.00	Rs 608	Rs 60,800	2.0%
Urea	4.9	92.00	Rs 454	Rs 45,400	1.5%
Limestone	4.3	25.00	Rs 107	Rs 10,700	0.3%
Feed subtotal	—	—	Rs 30,785	Rs 3,078,500	100%
Minerals & additives	—	—	Rs 4,007	Rs 400,700	—
TOTAL	—	—	Rs 34,792	Rs 3,479,200	—

Weekly cost progression

Week	Weight (kg)	Gain (kg)	Cost Rs	Rs/kg gain	Cumulative
1	165–170	4.6	1,486	327	Rs 1,486
2	170–178	8.1	2,177	268	Rs 3,663
3	178–186	8.2	2,254	275	Rs 5,917

Week	Weight (kg)	Gain (kg)	Cost Rs	Rs/kg gain	Cumulative
4	186–194	8.3	2,347	282	Rs 8,264
5	194–203	8.4	2,425	289	Rs 10,689
6	203–211	8.5	2,505	296	Rs 13,194
7	211–220	8.5	2,586	303	Rs 15,780
8	220–228	8.6	2,668	310	Rs 18,448
9	228–237	8.7	2,751	317	Rs 21,199
10	237–246	8.8	2,835	324	Rs 24,034
11	246–254	8.8	2,903	332	Rs 26,937
12	254–263	8.8	2,987	339	Rs 29,924
13	263–272	8.9	3,073	346	Rs 32,997
14	272–277	5.1	1,795	353	Rs 34,792
Total	165–277	112	34,792	310	Rs 34,792

Compared with the baseline, this scenario produces **12 kg more animal** over the same 95 days and lowers the all-in cost of gain from Rs 320 to Rs 310. Total cost per head rises from about Rs 32,000 to Rs 34,800 — but you are buying 112 kg of gain instead of 100 kg. The extra 12 kg costs roughly Rs 235 per kilogram, well below any plausible selling price.

16B. Scenario B — intake 3.1% declining to 2.9% of body weight

This scenario assumes cattle consume 3.1% of body weight in dry matter at 160 kg, declining to 2.9% at 280 kg, before the monsoon reduction. This is toward the upper end of what well-managed cattle achieve and would require excellent bunk management, ample trough space, shade and water.

Measure	Value
Purchase weight	165 kg
Days on feed	95
Exit weight	289.0 kg
Total gain	124.0 kg
Average daily gain	1.31 kg/day
Feed cost per head	Rs 33,588
Minerals and additives	Rs 4,009
Total cost per head	Rs 37,597
ALL-IN COST PER KG OF GAIN	Rs 303

Weekly ration — kilograms as fed per head per day

Wk	Live wt (kg)	Silage	Maize grain	Gluten feed	Toori	Urea	Limestone	ADG	Rs/day	Rs/week
1	165–170	5.90	1.70	0.12	0.27	0.028	0.024	0.65	220	1,539
2	170–178	6.10	2.90	0.20	0.26	0.047	0.041	1.28	329	2,303
3	178–188	6.52	3.05	0.20	0.23	0.049	0.043	1.30	344	2,405
4	188–197	6.97	3.20	0.20	0.20	0.052	0.045	1.32	359	2,510
5	197–206	7.19	3.33	0.22	0.24	0.054	0.047	1.33	372	2,601
6	206–216	7.42	3.47	0.23	0.27	0.056	0.048	1.34	385	2,693
7	216–225	7.98	3.62	0.23	0.20	0.058	0.050	1.36	400	2,802
8	225–235	8.26	3.76	0.24	0.21	0.061	0.052	1.37	414	2,897
9	235–244	8.57	3.90	0.24	0.22	0.063	0.054	1.38	428	2,993
10	244–254	8.90	4.04	0.25	0.22	0.065	0.056	1.39	441	3,090
11	254–264	9.26	4.18	0.26	0.21	0.067	0.058	1.40	455	3,188
12	264–274	9.28	4.30	0.28	0.30	0.070	0.060	1.40	467	3,269
13	274–283	9.31	4.42	0.30	0.39	0.072	0.062	1.40	478	3,348
14	283–289	9.35	4.54	0.33	0.47	0.074	0.064	1.40	490	1,959

Total feed consumed and cost per head over the 95-day cycle

Item	kg per head	Rs/kg	Cost per head	Per 100 head	% of feed cost
Maize silage	764.3	16.25	Rs 12,420	Rs 1,242,000	37.0%
Maize grain	334.3	55.00	Rs 18,387	Rs 1,838,700	54.7%
Maize gluten feed 30%	21.8	68.00	Rs 1,485	Rs 148,500	4.4%
Toori	24.4	28.00	Rs 682	Rs 68,200	2.0%
Urea	5.4	92.00	Rs 497	Rs 49,700	1.5%
Limestone	4.7	25.00	Rs 117	Rs 11,700	0.3%
Feed subtotal	—	—	Rs 33,588	Rs 3,358,800	100%
Minerals & additives	—	—	Rs 4,009	Rs 400,900	—
TOTAL	—	—	Rs 37,597	Rs 3,759,700	—

Weekly cost progression

Week	Weight (kg)	Gain (kg)	Cost Rs	Rs/kg gain	Cumulative
1	165–170	4.6	1,539	338	Rs 1,539
2	170–178	9.0	2,303	257	Rs 3,842
3	178–188	9.1	2,405	264	Rs 6,247

Week	Weight (kg)	Gain (kg)	Cost Rs	Rs/kg gain	Cumulative
4	188–197	9.2	2,510	272	Rs 8,757
5	197–206	9.3	2,601	279	Rs 11,358
6	206–216	9.4	2,693	287	Rs 14,051
7	216–225	9.5	2,802	294	Rs 16,853
8	225–235	9.6	2,897	302	Rs 19,750
9	235–244	9.7	2,993	310	Rs 22,743
10	244–254	9.7	3,090	318	Rs 25,833
11	254–264	9.8	3,188	325	Rs 29,021
12	264–274	9.8	3,269	334	Rs 32,290
13	274–283	9.8	3,348	342	Rs 35,638
14	283–289	5.6	1,959	350	Rs 37,597
Total	165–289	124	37,597	303	Rs 37,597

This is the strongest plan in output terms. It yields **124 kg of gain against the baseline's 100 kg — 24% more animal from the same pen, the same labour and the same 95 days** — while lowering the all-in cost of gain from Rs 320 to Rs 303. The additional 24 kg of gain costs about Rs 234 per kilogram at the margin.

Do not adopt this plan on assumption. Feeding to a 3.1% intake target when the animals will only eat 2.7% simply leaves expensive feed in the khurli to spoil in monsoon heat, and encourages sorting. Measure intake first using the protocol in the main report — from day 14 onward, weighing offered feed against refusals, targeting 2–5% refusal — and adopt this schedule only if the measurement supports it.

16C. Scenario C — green chari available at Rs 8 per kilogram

This scenario assumes fresh green chari can be bought at Rs 8/kg as fed, at the conservative baseline intake. Chari is about 22% dry matter, so it substitutes for silage rather than for grain, and is capped at 30% of dry matter. Silage falls to roughly 0.6–1.5 kg per day and chari carries the bulk of the forage.

Measure	Value
Purchase weight	165 kg
Days on feed	95
Exit weight	265.1 kg
Total gain	100.1 kg
Average daily gain	1.05 kg/day
Feed cost per head	Rs 24,770
Minerals and additives	Rs 4,011
Total cost per head	Rs 28,781
ALL-IN COST PER KG OF GAIN	Rs 288

Weekly ration — kilograms as fed per head per day

Wk	Live wt (kg)	Silage	Chari	Maize grain	Gluten feed	Toori	Urea	Limestone	ADG	Rs/day	Rs/week
1	165–170	0.78	5.45	1.48	0.11	0.21	0.024	0.018	0.65	189	1,320
2	170–177	0.63	5.58	2.51	0.20	0.27	0.041	0.030	1.03	262	1,835
3	177–184	0.91	5.80	2.62	0.18	0.20	0.043	0.031	1.05	273	1,908
4	184–192	0.97	6.01	2.72	0.19	0.20	0.044	0.033	1.06	281	1,966
5	192–199	1.02	6.22	2.82	0.20	0.20	0.046	0.034	1.07	290	2,028
6	199–206	0.83	6.44	2.90	0.22	0.28	0.047	0.035	1.07	296	2,073
7	206–214	0.93	6.65	3.00	0.22	0.26	0.049	0.036	1.08	305	2,135
8	214–222	1.04	6.86	3.10	0.23	0.25	0.050	0.037	1.09	314	2,197
9	222–229	1.18	7.07	3.20	0.23	0.22	0.052	0.038	1.10	323	2,260
10	229–237	1.02	7.28	3.28	0.24	0.29	0.053	0.039	1.10	330	2,308
11	237–245	1.19	7.48	3.38	0.24	0.25	0.055	0.040	1.11	339	2,372
12	245–253	1.38	7.69	3.49	0.24	0.21	0.056	0.042	1.12	348	2,436
13	253–261	1.25	7.89	3.57	0.26	0.27	0.058	0.043	1.12	355	2,485
14	261–265	1.48	8.10	3.67	0.25	0.22	0.059	0.044	1.13	364	1,458

Total feed consumed and cost per head over the 95-day cycle

Item	kg per head	Rs/kg	Cost per head	Per 100 head	% of feed cost
Maize silage	128.4	16.25	Rs 2,086	Rs 208,600	8.4%
Maize grain	276.8	55.00	Rs 15,225	Rs 1,522,500	61.5%
Maize gluten feed 30%	20.0	68.00	Rs 1,358	Rs 135,800	5.5%
Toori	22.6	28.00	Rs 632	Rs 63,200	2.6%
Urea	4.5	92.00	Rs 413	Rs 41,300	1.7%
Limestone	3.3	25.00	Rs 83	Rs 8,300	0.3%
chari	620.9	8.00	Rs 4,967	Rs 496,700	20.1%
Feed subtotal	—	—	Rs 24,770	Rs 2,477,000	100%
Minerals & additives	—	—	Rs 4,011	Rs 401,100	—
TOTAL	—	—	Rs 28,781	Rs 2,878,100	—

Weekly cost progression

Week	Weight (kg)	Gain (kg)	Cost Rs	Rs/kg gain	Cumulative
1	165–170	4.6	1,320	290	Rs 1,320
2	170–177	7.2	1,835	255	Rs 3,155

Week	Weight (kg)	Gain (kg)	Cost Rs	Rs/kg gain	Cumulative
3	177–184	7.4	1,908	260	Rs 5,063
4	184–192	7.4	1,966	265	Rs 7,029
5	192–199	7.5	2,028	271	Rs 9,057
6	199–206	7.5	2,073	277	Rs 11,130
7	206–214	7.6	2,135	282	Rs 13,265
8	214–222	7.6	2,197	288	Rs 15,462
9	222–229	7.7	2,260	293	Rs 17,722
10	229–237	7.7	2,308	300	Rs 20,030
11	237–245	7.8	2,372	305	Rs 22,402
12	245–253	7.8	2,436	311	Rs 24,838
13	253–261	7.8	2,485	317	Rs 27,323
14	261–265	4.5	1,458	323	Rs 28,781
Total	165–265	100	28,781	288	Rs 28,781

Chari at Rs 8 is the cheapest plan per kilogram of gain — **Rs 288 against the baseline Rs 320, a saving of about Rs 3,200 per head or Rs 320,000 across 100 head**. Output is unchanged at 100 kg of gain and a 265 kg exit; the saving is purely on cost.

Cautions. Chari is seasonal — cutting generally runs June to September and again in spring — so it cannot replace silage year round, and your silage remains necessary for the months chari is unavailable. Chari at above about Rs 14/kg is not worth buying. Introduce it gradually over 5–7 days rather than switching in one day, and never feed chari cut young or after drought stress without checking for nitrate and prussic acid risk, which is a genuine hazard in sorghum-type fodder.

Note also that the minimum-gain floor of 1.00 kg/day is doing real work in this scenario. Without it, the optimiser would feed far more chari and settle at about 0.76 kg/day, exiting near 237 kg and missing the 95-day window. Cheap bulky feed must always be paired with a minimum gain constraint.

17. How to read the numbers — the two cost bases

Two different cost bases appear across these documents and they are not interchangeable. Mixing them up is easy, so both are defined here.

Term	What it means	Where it appears	Example
Blended feed-only cost per kg gain	The ration is solved at five body weights (160, 190, 220, 250 and 280 kg) and the resulting cost per kilogram of gain is averaged. It counts feed ingredients only — no minerals, no additives, no adaptation week.	The intake and dry-matter comparison tables (Sections 10C and 10D of the main report)	Rs 257 at 3.1% intake and 30% DM
All-in cost per kg gain over the full cycle	A day-by-day simulation of all 95 days, including the week-1 adaptation period at reduced gain, plus the mineral, vitamin, buffer and binder pack at Rs 43.8 per head per day. Total cost divided by total gain.	The headline figures and every weekly schedule in this supplement	Rs 303 for the same 3.1% intake scenario

Why the all-in figure is higher. Three things are added. First, the mineral and additive pack costs Rs 43.8 per head per day, which adds roughly Rs 40 per kilogram of gain. Second, week 1 is an adaptation week in which the animal gains only about 0.65 kg/day while still eating, so its cost per kilogram is high. Third, the cycle simulation spends more of its time at heavier weights, where cost per kilogram of gain is naturally higher, than the simple five-point average does.

So for the same scenario you will see roughly **Rs 257 feed-only** and **Rs 303 all-in**. Neither is wrong; they answer different questions. Use the **feed-only** figures when comparing ingredients, prices or scenarios against each other, because they isolate the effect you are testing. Use the **all-in** figures for budgeting, cash-flow planning and margin calculations, because that is what you will actually spend.

A worked example of a single cell

The cell reading **Rs 257 | 1.33 kg/d | exit 291 kg** under the 30% dry matter column and the 3.1% intake row means: if your silage tests at 30% dry matter *and* your animals eat 3.1% of body weight in dry matter, then the least-cost ration costs Rs 257 in feed ingredients for every kilogram of live weight the animal gains; the animal gains an average of 1.33 kg per day across the cycle; and starting from 165 kg it would reach about 291 kg after 95 days. Adding the mineral pack and the adaptation week brings the true budgeting figure to roughly Rs 295 per kilogram.

One further caution on these tables. Each cell is a hypothetical. The intake rows of 2.9% and 3.1% are not measurements of your animals — they are what the model produces *if* your animals eat that much. Likewise the dry-matter columns are not measurements of your silage. Until both are verified, the only cell describing your farm as it stands is the baseline: 2.7% intake at 27% dry matter.

18. Comparison — where the money goes

Feed cost per head across the four plans, by ingredient:

Ingredient	Baseline	A — intake 2.9%	B — intake 3.1%	C — chari Rs 8
Maize silage	Rs 9,409	Rs 11,441	Rs 12,420	Rs 2,086
chari	—	—	—	Rs 4,967
Maize grain	Rs 15,241	Rs 16,822	Rs 18,387	Rs 15,225
Maize gluten feed 30%	Rs 1,217	Rs 1,349	Rs 1,485	Rs 1,358
Toori	Rs 532	Rs 608	Rs 682	Rs 632
Urea	Rs 414	Rs 454	Rs 497	Rs 413
Limestone	Rs 98	Rs 107	Rs 117	Rs 83
Feed subtotal	Rs 26,911	Rs 30,785	Rs 33,588	Rs 24,770
Minerals & additives	Rs 4,003	Rs 4,007	Rs 4,009	Rs 4,011
TOTAL PER HEAD	Rs 31,969	Rs 34,792	Rs 37,597	Rs 28,781
Gain produced	100 kg	112 kg	124 kg	100 kg
ALL-IN Rs PER KG GAIN	Rs 320	Rs 310	Rs 303	Rs 288

Maize grain remains the largest single line in every plan, which is why harvest-time procurement matters more than any of these choices. Note that the higher-intake plans consume more of everything — they are not cheaper in total, they are more productive. The chari plan is the only one that reduces total spending, and it does so by displacing silage.

19. Which plan to pursue

Priority	Action	Why
1	Measure actual intake from day 14	Determines whether Scenario A or B is real. Costs nothing and is worth up to 24% more output.
2	Confirm chari price and availability	Scenario C saves Rs 320,000 across 100 head if chari can be had at Rs 8. Currently out of season.
3	Confirm silage dry matter by microwave test	Every percentage point is worth Rs 3–4 per kg of gain.
4	Secure maize grain at the October–November harvest	Still the single largest lever available, worth more than all three scenarios combined.

These scenarios are not mutually exclusive. If intake proves high *and* chari becomes available in season, both effects apply together and the combined figure would be lower than either alone. That combination has not been modelled here because both inputs remain unverified.

20. What would sharpen the model further

Priority	Item	Why
1	Confirm khal and maize oil cake bag sizes (40 or 50 kg)	Low impact on cost (about Rs 2) but closes the last open price question.
2	Actual chari price and availability window	Section 10 shows a saving of roughly Rs 34/kg gain at Rs 8. Worth confirming whether it can be sourced.
3	Rafhan corn steep liquor and hydrol prices	Enter the ration below Rs 43.5 and Rs 24 respectively.
4	Bakery waste and cull potato quotes	Break-evens of Rs 78 and Rs 14.4 — both realistically achievable.
5	Ionophore price and confirmation it is being fed	A 6% efficiency gain is assumed but the additive is not costed.
6	Mineral premix price and guaranteed analysis	Currently estimated at Rs 400/kg.
7	Observed intake — do animals clean up the silage quantity in Section 3?	If they consistently eat more, the intake ceiling is conservative and both gain and silage can rise.
8	Silage DM confirmed by microwave test at feed-out	Currently taken as 27%. Each point is worth Rs 4–5 per kg of gain.

21. Assumptions and limitations

- Feed composition is from standard published tables, not laboratory analysis of your feeds. Maize gluten feed is modelled at 30% protein as fed; mithi khal at 22% and kori khal at 19%.
- Maize silage is taken as 27% dry matter and assumed fully fermented (21+ days). Limestone price is an estimate.
- Khal and maize oil cake bag sizes are assumed to be 50 kg.
- A genetic ceiling of about 1.2 kg/day is assumed for the Sahiwal × Friesian cross. Achieved figures are lower because of monsoon heat and the adaptation week.
- The model minimises cost per kg of gain subject to a minimum daily gain of 1.00 kg. Without this floor the optimiser selects cheaper, slower growth that misses the 95-day window.
- Monsoon adjustment applies +10% to maintenance energy and –10% to intake, representing competent management at THI 80–88. Poor heat management roughly halves gain.
- Maintenance energy is reduced 7% for zebu content. An ionophore benefit of 6% is assumed and the ionophore is not separately costed.
- Break-even prices are computed at 220 kg body weight and shift slightly at other weights.
- Costs cover feed and additives only. Labour, veterinary care, vaccination, mortality, transport, financing and the animal's purchase price are excluded.
- Gain is modelled as a smooth average; real animals vary. Sort pens by weight and read the bunk daily.
- This is a planning model, not veterinary advice. Have a qualified nutritionist or veterinarian approve urea, ionophore and mineral levels before herd-wide use.

22. Sources

1. NASEM (2016). Nutrient Requirements of Beef Cattle, 8th revised edition. National Academies Press — net energy equations.
2. Merck Veterinary Manual (2026). Nutrient Requirements of Beef Cattle; Urea Poisoning in Ruminants; Urolithiasis in Ruminants.
3. Oklahoma State University Extension E-974; Mississippi State University Extension — requirement tables adapted from NASEM/NRC.
4. Galyean M. & Hales K. et al. (2022). Metabolizable energy prediction for growing and finishing cattle. PMC8892684.
5. Farooq K. et al. (2024). Optimising maize silage in forage-blend rations for fattening calves. Int. J. Agricultural and Biological Engineering 17(3):75–81.
6. Standard feed composition tables (FAO feed classification; NRC feed library) for maize grain and byproducts, cottonseed products, brans, molasses, straw and NPN sources.
7. Silage fermentation phases and aerobic stability — standard forage conservation literature.
8. Temperature-humidity index thresholds and heat-stress management for cattle — standard extension guidance.
9. All feed prices supplied by BA Farms, July 2026, except limestone which is an estimate as marked in Section 1.
10. Optimisation performed with SciPy linprog (HiGHS solver); source code accompanies this report.